


# Coupling Interpretable Allocation, Model-Predictive Control, and Tail-Aware Forecasting for Live Portfolio Management

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# Outline

Live portfolio management couples three problems:
allocation, the timing of rebalancing, and evaluation beyond the historical record.

| | | |
|---|---|---|
| 1 | **Allocation** | **What to hold.** Maximize utility of a score γ for each stock, within trading limits. |
| 2 | **Rebalancing** | **When to trade.** Re-plan at the 21-day horizon, a band exit, or a drawdown stop; real-data backtest, 2025–26. |
| 3 | **Market generator** | **Evaluation.** A backtest is one historical path; JumpHMM generates typical and stressed futures. |

# Allocation: maximize utility subject to trading limits

Each rebalance chooses $n_i$ shares of each name $i$ to solve:

$$\max_{n \geq 0} \; U(n;\gamma)$$ — utility $U$ ($\gamma_i$: score of name $i$, next slide)

s.t. $\sum_i n_i S_i \leq W$ — budget $W$ ($S_i$: price of name $i$)

$w_i \leq \bar{w}$ — single-name cap $\bar{w}$ ($w_i$: weight of name $i$)

$\sqrt{w^\top \Sigma_g \, w} \leq \sigma_{\text{SPY}}$ — volatility cap $\sigma_{\text{SPY}}$ ($\Sigma_g$: growth-rate covariance)

$\bar{c} \, \|n - n^{\text{prev}}\|_1 \leq B$ — turnover budget $B$ ($\bar{c}$: half spread per share)

$\beta^\top w \leq \bar{\beta}$ — β cap $\bar{\beta}$, hedge on (β: beta to SPY)

| | **Cobb–Douglas** | **CES on shares** | **CES on weights** | **Stone–Geary** |
|---|---|---|---|---|
| $U(n;\gamma)$, over $\gamma_i > 0$ | $\sum \gamma_i \ln n_i$ | $(\sum \gamma_i n_i^\rho)^{1/\rho}$ | $(\sum \gamma_i w_i^\rho)^{1/\rho}$ | $\sum \gamma_i \ln(n_i - b_i)$ |
| budget alone | $w_i \propto \gamma_i$ | $n_i \propto (\gamma_i / S_i)^\eta$ | $w_i \propto \gamma_i^\eta$ | $(n_i - b_i)\, S_i \propto \gamma_i$ |
| with the limits | a convex cone program for each utility, solved numerically | | | |

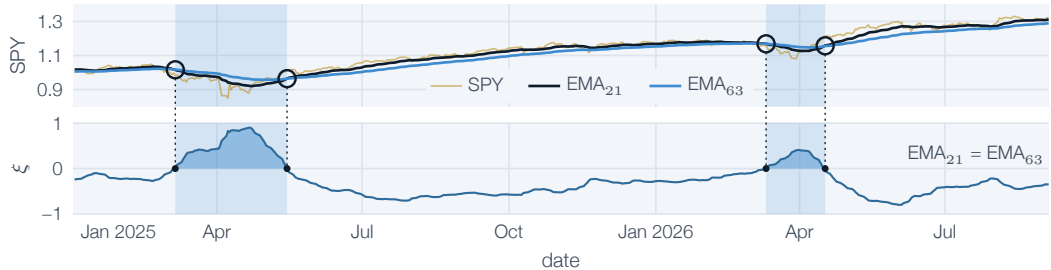
$\rho = 1 - 1/\eta$, $\eta = 0.5$; $b_i S_i = 1\%$ of $W$. Benchmarks under the same limits: minimum variance, tangency, 130/30.

# The score γ: expected growth, tilted by the market

$$\gamma_i = \tanh\left(\frac{\alpha_i + \beta_i\,\tilde{g}_M}{\max(|\beta_i|,\,0.1)^{\xi}}\right)$$

expected growth ($\alpha_i, \beta_i$: fit of name $i$ to SPY;
$\tilde{g}_M$: SPY's growth rate, 252-day EMA)

β penalty ($\xi$: market state, > 0 when $\text{EMA}_{21} < \text{EMA}_{63}$)



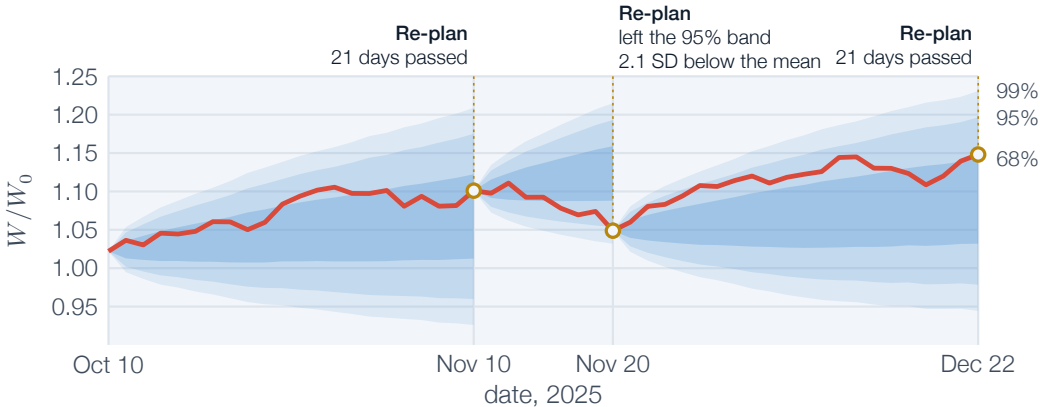

**Weighting** γ enters $U(n;\gamma)$: names with $\gamma_i > 0$ are held long, and 130/30 shorts $\gamma_i < 0$.

**Selection** the 13 highest γ overall, or the 3 highest in each sector.

$\xi = -G\,(\text{EMA}_{21}/\text{EMA}_{63} - 1)$ from SPY's 21- and 63-day exponential averages; $|\xi| \le 1$ on 99% of 2014–2024 days.

# Rebalancing triggers: horizon, forecast band, drawdown

**Forecast:** multi-asset GBM, 1,000 paths, 21 trading days ahead. **Plant:** real prices.



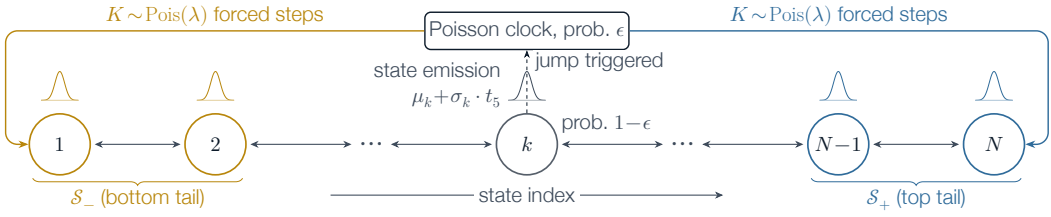

Hand-picked 13, Cobb–Douglas, hedge off; over Jan 2025 – Sep 2026, 27 re-plans, 10 before the horizon.

# Stock selection matters more than the weighting rule

Real-data backtest, Jan 2025 – Sep 2026, 373 names. Each row is a selection rule.



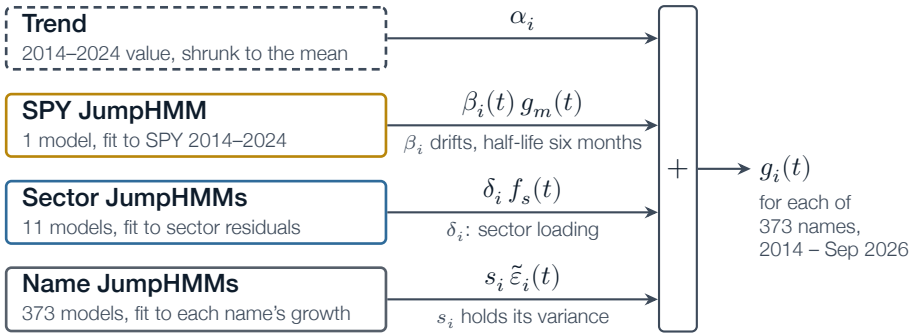
# JumpHMM: a market generator with fat tails




The controller forecasts with GBM; JumpHMM generates the test markets. Alswaidan and Varner, arXiv:2603.10202.

# From one market to 373 names

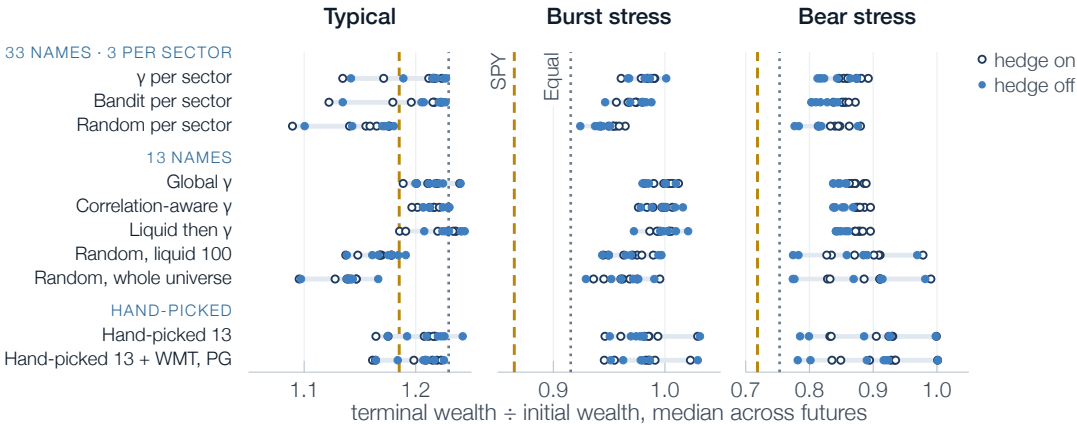
Three kinds of JumpHMM, composed into each name's daily growth rate.




Residuals are not reordered: each keeps its own clustering and jump timing.

# Simulated sell-offs: every median ends above SPY

600 typical JumpHMM futures of Jan 2025 – Sep 2026, and 200 in each stress set.




Stress sets: of 10,000 paths, the 200 deepest SPY drawdowns with a jump burst, and the 200 without one.

# Key takeaways

- **Allocation: stock selection mattered more than the weighting rule.** With the hedge off, seven weighting rules on one 33-name basket ended at 1.09–1.23 times initial wealth; different baskets ended at 1.03–1.70.
- **Rebalancing: re-plans came at the horizon, a band exit, or a drawdown stop.** About 27 re-plans in 419 trading days, a third of them before the 21-day horizon.
- **Evaluation: in both sell-off sets, every median ended above SPY.** JumpHMM reproduces fat tails, but its volatility clustering fades after a day. With the hedge off, 13 names picked by γ beat 13 random names in 62–71% of typical futures; in bear futures, random picks did as well or better for most rules.

A paper-trading version of this system is running now; its results are not part of this talk.

# Acknowledgements

**Abdulrahman Alswaidan**, MS 2026
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helpful discussions

**Evan Vu**, class of 2026, now at Bridgewater
an early version of the Cobb–Douglas allocation and the bandit picker

We also thank Alpaca Markets.

**Paper**



JumpHMM paper
arXiv:2603.10202

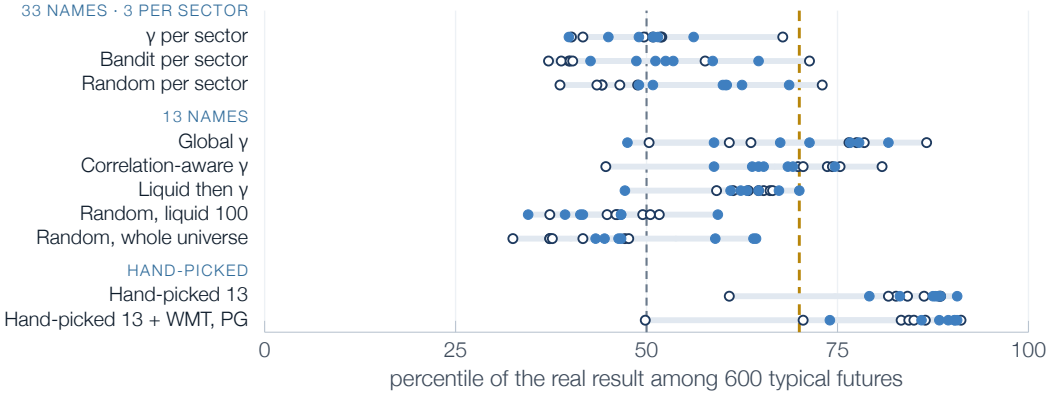
**Code**



github.com/varnerlab/
INFORMS-Annual-
Meeting-2026-Varnerlab

# Real results as percentiles of typical futures

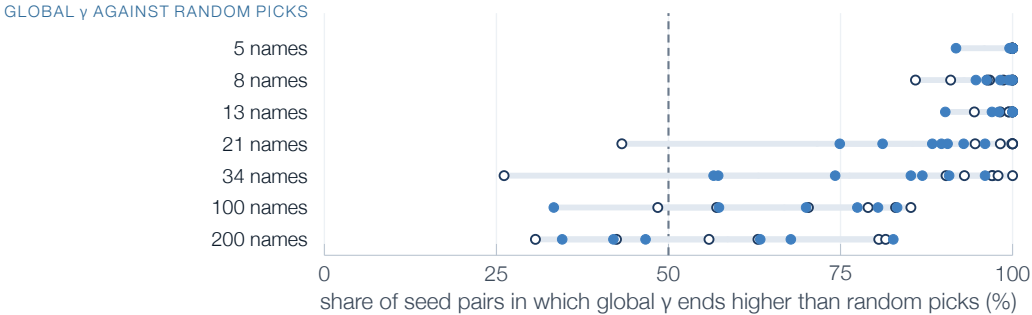
Each real result among 600 typical JumpHMM futures. Right of 50 beats the median future.




Real results use the simulated runs' seeds: 5660–5664 for random and bandit pickers, 5660 for the rest.

# Global γ against random picks, 5 to 200 names

The K highest γ in the universe against K random names, from 5 to 200 names. At 50%, the two picks end higher equally often.




● hedge off ○ hedge on one dot per weighting rule

Real-data backtest, 30 runs each. Stone–Geary left out at 100 and 200 names: its 1%-per-name floor uses the whole budget at 100 names and twice the budget at 200.

# Terminal wealth by picker and utility: hedge off

| Picker | CD | CES | CESw | SG | MinVar | Tangent | LS |
|---|---|---|---|---|---|---|---|
| γ per sector | 1.230 | 1.219 | 1.221 | 1.234 | 1.086 | 1.227 | 1.110 |
| Bandit per sector | 1.226 | 1.234 | 1.223 | 1.239 | 1.162 | 1.229 | 1.140 |
| Random per sector | 1.205 | 1.183 | 1.201 | 1.183 | 1.128 | 1.158 | 1.127 |
| Global γ | 1.520 | 1.384 | 1.460 | 1.441 | 1.355 | 1.341 | 1.248 |
| Correlation-aware γ | 1.352 | 1.386 | 1.421 | 1.380 | 1.284 | 1.404 | 1.219 |
| Liquid then γ | 1.333 | 1.310 | 1.379 | 1.350 | 1.238 | 1.361 | 1.305 |
| Random, liquid 100 | 1.216 | 1.173 | 1.211 | 1.212 | 1.158 | 1.274 | 1.152 |
| Random, whole universe | 1.098 | 1.132 | 1.138 | 1.112 | 1.104 | 1.160 | 1.031 |
| Hand-picked 13 | 1.548 | 1.571 | 1.614 | 1.631 | 1.443 | 1.697 | 1.549 |
| Hand-picked 13 + WMT, PG | 1.616 | 1.646 | 1.637 | 1.639 | 1.362 | 1.642 | 1.622 |
| SPY buy-and-hold | 1.317 | 1.317 | 1.317 | 1.317 | 1.317 | 1.317 | 1.317 |
| equal-weight universe | 1.296 | 1.296 | 1.296 | 1.296 | 1.296 | 1.296 | 1.296 |

Terminal wealth ÷ initial wealth, median of 30 runs. Real-data backtest, Jan 2025 – Sep 2026.
CD Cobb–Douglas · CES on shares · CESw CES on weights · SG Stone–Geary · MinVar minimum variance · Tangent the tangency portfolio · LS 130/30 long-short Cobb–Douglas.

# Terminal wealth by picker and utility: hedge on

| Picker | CD | CES | CESw | SG | MinVar | Tangent | LS |
|---|---|---|---|---|---|---|---|
| γ per sector | 1.212 | 1.232 | 1.223 | 1.218 | 1.073 | 1.285 | 1.130 |
| Bandit per sector | 1.125 | 1.140 | 1.167 | 1.194 | 1.166 | 1.214 | 1.120 |
| Random per sector | 1.108 | 1.087 | 1.114 | 1.155 | 1.155 | 1.092 | 1.092 |
| Global γ | 1.503 | 1.305 | 1.464 | 1.461 | 1.203 | 1.589 | 1.283 |
| Correlation-aware γ | 1.409 | 1.429 | 1.479 | 1.447 | 1.213 | 1.366 | 1.292 |
| Liquid then γ | 1.322 | 1.285 | 1.347 | 1.331 | 1.184 | 1.317 | 1.336 |
| Random, liquid 100 | 1.167 | 1.156 | 1.174 | 1.178 | 1.123 | 1.155 | 1.112 |
| Random, whole universe | 1.069 | 1.106 | 1.093 | 1.115 | 1.079 | 1.062 | 0.994 |
| Hand-picked 13 | 1.554 | 1.517 | 1.517 | 1.511 | 1.233 | 1.582 | 1.586 |
| Hand-picked 13 + WMT, PG | 1.564 | 1.571 | 1.543 | 1.542 | 1.159 | 1.383 | 1.630 |
| SPY buy-and-hold | 1.317 | 1.317 | 1.317 | 1.317 | 1.317 | 1.317 | 1.317 |
| equal-weight universe | 1.296 | 1.296 | 1.296 | 1.296 | 1.296 | 1.296 | 1.296 |

Terminal wealth ÷ initial wealth, median of 30 runs. Real-data backtest, Jan 2025 – Sep 2026.
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